

K SHREEKRISHNA UPADHYAYA

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PROFILE

AI/ML engineering student with hands-on experience in speech processing, deep learning, and AI-assisted software engineering. Applying strong software engineering fundamentals alongside machine learning to build reliable software systems

EDUCATION

VIVEKANANDA COLLEGE OF ENGINEERING AND TECHNOLOGY, PUTTUR

DK, Karnataka

Bachelor of Engineering in Artificial Intelligence & Machine Learning

September 2023 – Present

Cumulative GPA (till 6th semester): 8.69/10.0

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures

PROJECTS

DADS (Detection and Analysis of Dysfluencies in Speech) | Python, PyTorch, Librosa

September 2025 – November 2025

Stuttering dysfluency detection system on the SEP-28k dataset, exploring both per-type and multi-label CNN architectures.

- Trained 5 binary CNN classifiers (one per dysfluency type: prolongation, block, sound repetition, word repetition, interjection) on mel-spectrogram features with per-type hyperparameter search across window size, hop length, and mel bands
- Explored a CNN-LSTM multi-label classifier as an alternative to separate per-type models
- Built a PyQt5 desktop application with real-time audio recording, playback, spectrogram visualization, and background stutter detection inference

The Major Project extends this work with a Wav2Vec2 + CNN-transformer hybrid architecture.

FinDiary | Go, PostgreSQL, Flutter, gRPC

July 2026 – Present

Ongoing personal finance tracking application developed entirely through Specification-Driven Development — specifications are authored and refined before an agentic AI workflow generates implementation code, with all output reviewed for architectural consistency.

- Architecting Go backend services with PostgreSQL persistence and gRPC-based client-server communication
- Directing development of a Flutter cross-platform client, with offline-first local state management planned
- Practicing spec-first, AI-assisted implementation as a deliberate alternative to traditional hand-written development, alongside the fully hand-written systems work below

Sn Project | C, x86_64 Assembly

December 2025 – Present

Collection of 13 cross-platform C libraries (memory allocators, threading, logging, tracing) extracted from an experimental game engine into reusable, independently-versioned infrastructure.

- Designed 7 memory allocator strategies behind a unified interface; hand-wrote x86_64 atomics with explicit memory-ordering fences for cross-platform threading primitives

Snuk | C

April 2026 – Present

Embeddable scripting language with a custom lexer, recursive-descent parser, and tree-walking interpreter (6K lines of C), including reference-counted memory management.

hash shell | C

January 2026 – March 2026

Open-source contributor with 17 merged pull requests, including a self-proposed multi-PR refactoring initiative to align the codebase with upstream coding standards.

Additional Projects:

Regex Engine (Thompson NFA construction in C), StateFlow (DFA/NFA simulator with automaton validity checking), LC3VM (LC-3 virtual machine).

SKILLS

AI/ML: PyTorch, Transformers, Wav2Vec2, CNN Architectures, Model Fine-Tuning, Dataset Preprocessing & Evaluation, Specification-Driven / Agentic AI Workflows

Languages & Build: Python, C, C++, Go, Java, CMake, x86_64 Assembly, POSIX Shell

Development: Git, Linux, CI/CD (GitHub Actions), gRPC, REST APIs, PostgreSQL, Flutter, Typst

Achievements: Winner of Nexathon 2025 (SDIT) and DevHack 2025 (Sahyadri College)